

Method of Moments Solver for Thin-Wire Antennas

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Abstract—A Method of Moments (MoM) solver is developed for computing antenna current distributions and input impedances. The solver is validated on a delta-gap-fed half-wave dipole antenna and then applied to a ninety-degree bent-wire antenna to examine nonparallel segment coupling.

Index Terms—Method of moments, thin-wire EFIE, delta-gap source, Galerkin testing, triangular basis functions, bent-wire antenna.

I. INTRODUCTION

THIN-WIRE antennas are a standard class of problems in antenna analysis because their dominant unknown can be represented by a scalar current flowing along a conducting centerline. This reduces the electromagnetic problem from a volumetric current distribution to a one-dimensional wire-current problem. For straight dipoles and connected wire structures, the main quantities of interest are the current distribution along the wire and the input impedance at the feed. These quantities determine how the antenna radiates and how it couples to a source or transmission line [1].

The Method of Moments (MoM) is a direct numerical method for converting an integral equation into a finite-dimensional linear system. In Harrington's moment-method framework, the unknown function is expanded in basis functions, the residual is tested against weighting functions, and the resulting matrix equation relates expansion coefficients to the excitation vector [2]. For a wire antenna, the unknown coefficients represent samples or amplitudes of the line current, while each matrix entry represents the electromagnetic interaction between a testing function and a source basis function.

This report follows the thin-wire conducting-wire formulation in Section 10.3.2 of Jin's textbook [3]. The electric-field integral equation is written using the vector and scalar potentials. Because the scalar-potential contribution contains the derivative of the unknown current, triangular basis functions are used for the current expansion. The same triangular functions are used for Galerkin testing. After integration by parts, the matrix separates into a vector-potential contribution and a scalar-potential contribution. The vector-potential term contains the tangent projection between wire segments, while the scalar-potential term does not.

The validation case is a straight center-fed dipole of length 0.5λ and radius 0.001λ , excited by an ideal delta-gap voltage source. The numerical input impedance is compared with the approximate reference value $73 + j42.5 \Omega$, and several discretization densities are used to examine convergence behavior. As an additional example, a 90-degree bent-wire antenna is analyzed to demonstrate that the same solver applies when wire segments are not parallel.

The remainder of this paper is organized as follows. Section II derives the thin-wire EFIE from the frequency-domain potential representation. Section III defines the triangular current expansion, and Section IV gives the Galerkin-tested matrix equation. Section V describes the segment-pair evaluation, while Section VI discusses the overlapping-subsegment and ψ -integral treatment. Section VII defines the delta-gap source and input-impedance formula. Sections VIII and IX present the validation geometries and numerical results.

II. THIN-WIRE EFIE

Assume time dependence $e^{j\omega t}$. The free-space wavenumber and intrinsic impedance are [1], [3]

$$k_0 = \omega\sqrt{\mu_0\epsilon_0}, \quad Z_0 = \sqrt{\frac{\mu_0}{\epsilon_0}}. \quad (1)$$

The outgoing scalar Green function is [2], [3]

$$G_0(\mathbf{r}, \mathbf{r}') = \frac{e^{-jk_0 R}}{4\pi R}, \quad R = \|\mathbf{r} - \mathbf{r}'\|. \quad (2)$$

Let C be the wire centerline and let l denote arclength. The thin-wire current is represented as a scalar line current directed along the local tangent [3]:

$$\mathbf{I}(l) = \hat{\mathbf{t}}(l)I(l), \quad \hat{\mathbf{t}}(l) = \frac{d\mathbf{r}(l)}{dl}, \quad (3)$$

where $I(l)$ is the scalar line current.

The scattered electric field is expressed through the frequency-domain potentials [2], [3]:

$$\mathbf{E}^{\text{sc}} = -j\omega\mathbf{A} - \nabla\Phi. \quad (4)$$

The vector potential is

$$\mathbf{A}(\mathbf{r}) = \mu_0 \int_C \hat{\mathbf{t}}(l')I(l')G_0(\mathbf{r}, \mathbf{r}(l')) dl'. \quad (5)$$

The line charge density follows from continuity:

$$\frac{dI(l')}{dl'} + j\omega\rho_l(l') = 0, \quad (6)$$

so that

$$\rho_l(l') = -\frac{1}{j\omega} \frac{dI(l')}{dl'} = \frac{j}{\omega} \frac{dI(l')}{dl'}. \quad (7)$$

Therefore, the scalar potential is [3]

$$\begin{aligned} \Phi(\mathbf{r}) &= \frac{1}{\epsilon_0} \int_C \rho_l(l')G_0(\mathbf{r}, \mathbf{r}(l')) dl' \\ &= \frac{j}{\omega\epsilon_0} \int_C \frac{dI(l')}{dl'} G_0(\mathbf{r}, \mathbf{r}(l')) dl' \\ &= \frac{jZ_0}{k_0} \int_C I'(l')G_0(\mathbf{r}, \mathbf{r}(l')) dl', \end{aligned} \quad (8)$$

where $I' = dI/dl'$.

On the perfectly conducting wire surface, the total tangential electric field vanishes [1], [3]:

$$\hat{\mathbf{t}}(l) \cdot [\mathbf{E}^{\text{inc}}(\mathbf{r}(l)) + \mathbf{E}^{\text{sc}}(\mathbf{r}(l))] = 0. \quad (9)$$

Using (4), (5), and (8), the tangential scattered field is

$$\begin{aligned} \hat{\mathbf{t}}(l) \cdot \mathbf{E}^{\text{sc}}(\mathbf{r}(l)) = & -jk_0 Z_0 \int_C \hat{\mathbf{t}}(l) \cdot \hat{\mathbf{t}}(l') I(l') G_0(\mathbf{r}(l), \mathbf{r}(l')) dl' \\ & - \frac{jZ_0}{k_0} \int_C I'(l') \frac{dG_0(\mathbf{r}(l), \mathbf{r}(l'))}{dl} dl'. \end{aligned} \quad (10)$$

Here the observation derivative is

$$\frac{dG_0(\mathbf{r}(l), \mathbf{r}(l'))}{dl} = \hat{\mathbf{t}}(l) \cdot \nabla_{\mathbf{r}} G_0(\mathbf{r}, \mathbf{r}(l')). \quad (11)$$

Substitution into the PEC boundary condition gives the thin-wire EFIE [3]:

$$\begin{aligned} jk_0 Z_0 \int_C [\hat{\mathbf{t}}(l) \cdot \hat{\mathbf{t}}(l') I(l') G_0(\mathbf{r}(l), \mathbf{r}(l')) \\ + \frac{1}{k_0^2} I'(l') \frac{dG_0(\mathbf{r}(l), \mathbf{r}(l'))}{dl}] dl' \\ = \hat{\mathbf{t}}(l) \cdot \mathbf{E}^{\text{inc}}(\mathbf{r}(l)). \end{aligned} \quad (12)$$

III. TRIANGULAR CURRENT EXPANSION

The wire is divided into N_s straight segments and $N_s + 1$ nodes. Segment p connects node $p - 1$ to node p , for $p = 1, 2, \dots, N_s$. Its length and unit tangent are

$$\Delta l_p = \|\mathbf{r}_p - \mathbf{r}_{p-1}\|, \quad (13)$$

$$\hat{\mathbf{t}}_p = \frac{\mathbf{r}_p - \mathbf{r}_{p-1}}{\Delta l_p}. \quad (14)$$

The arclength coordinate of node i is

$$s_0 = 0, \quad s_i = \sum_{p=1}^i \Delta l_p. \quad (15)$$

For an open wire, the endpoint currents are zero [3]:

$$I_0 = 0, \quad I_{N_s} = 0. \quad (16)$$

The unknown nodal currents are therefore

$$I_1, I_2, \dots, I_{N_s-1}. \quad (17)$$

The current is expanded using triangular basis functions [2], [3]:

$$I_h(l) = \sum_{n=1}^{N_s-1} I_n \Lambda_n(l). \quad (18)$$

The triangular basis function centered at node n is

$$\Lambda_n(s) = \begin{cases} \frac{s - s_{n-1}}{\Delta l_n}, & s_{n-1} \leq s \leq s_n, \\ \frac{s_{n+1} - s}{\Delta l_{n+1}}, & s_n \leq s \leq s_{n+1}, \\ 0, & \text{otherwise.} \end{cases} \quad (19)$$

Its derivative is piecewise constant:

$$\Lambda'_n(s) = \begin{cases} \frac{1}{\Delta l_n}, & s_{n-1} \leq s \leq s_n, \\ -\frac{1}{\Delta l_{n+1}}, & s_n \leq s \leq s_{n+1}, \\ 0, & \text{otherwise.} \end{cases} \quad (20)$$

IV. GALERKIN TESTING AND MATRIX EQUATION

Substituting (18) into (12) and testing with Λ_m gives the Galerkin MoM matrix equation [2], [3]:

$$\sum_{n=1}^{N_s-1} Z_{mn} I_n = V_m, \quad m = 1, 2, \dots, N_s - 1. \quad (21)$$

The excitation vector is

$$V_m = \int_C \Lambda_m(l) \hat{\mathbf{t}}(l) \cdot \mathbf{E}^{\text{inc}}(\mathbf{r}(l)) dl. \quad (22)$$

Before integration by parts, the impedance-matrix entry is [3]

$$\begin{aligned} Z_{mn} = & jk_0 Z_0 \int_C \int_C \Lambda_m(l) \Lambda_n(l') [\hat{\mathbf{t}}(l) \cdot \hat{\mathbf{t}}(l')] G_0(\mathbf{r}(l), \mathbf{r}(l')) dl' dl \\ & + \frac{jZ_0}{k_0} \int_C \int_C \Lambda_m(l) \Lambda'_n(l') \frac{dG_0(\mathbf{r}(l), \mathbf{r}(l'))}{dl} dl' dl. \end{aligned} \quad (23)$$

The derivative in the second term is with respect to the observation coordinate l . For fixed l' , integration by parts gives

$$\int_C \Lambda_m(l) \frac{dG_0}{dl} dl = [\Lambda_m(l) G_0]_{\partial C} - \int_C \Lambda'_m(l) G_0 dl. \quad (24)$$

Because Λ_m vanishes at the endpoints of the open wire, the boundary term is zero:

$$\int_C \Lambda_m(l) \frac{dG_0}{dl} dl = - \int_C \Lambda'_m(l) G_0 dl. \quad (25)$$

Therefore, the final Galerkin matrix entry is [3]

$$Z_{mn} = jk_0 Z_0 A_{mn} - \frac{jZ_0}{k_0} B_{mn}, \quad (26)$$

where

$$A_{mn} = \int_C \int_C \Lambda_m(l) \Lambda_n(l') [\hat{\mathbf{t}}(l) \cdot \hat{\mathbf{t}}(l')] G_0(\mathbf{r}(l), \mathbf{r}(l')) dl' dl, \quad (27)$$

$$B_{mn} = \int_C \int_C \Lambda'_m(l) \Lambda'_n(l') G_0(\mathbf{r}(l), \mathbf{r}(l')) dl' dl. \quad (28)$$

The tangent dot product appears only in the vector-potential contribution A_{mn} . It does not appear in the scalar-potential contribution B_{mn} , which is important for nonparallel wire geometries [3].

V. SEGMENT-PAIR EVALUATION

The support of Λ_m is the union of segments m and $m + 1$. Thus,

$$\mathcal{S}_m = \{m, m + 1\}, \quad \mathcal{S}_n = \{n, n + 1\}. \quad (29)$$

The impedance entry is evaluated by summing the four segment-pair interactions:

$$Z_{mn} = jk_0 Z_0 \sum_{p \in \mathcal{S}_m} \sum_{q \in \mathcal{S}_n} A_{mn}^{pq} - \frac{jZ_0}{k_0} \sum_{p \in \mathcal{S}_m} \sum_{q \in \mathcal{S}_n} B_{mn}^{pq}. \quad (30)$$

Segment p is parameterized by $\xi \in [0, 1]$:

$$\mathbf{r}_p(\xi) = \mathbf{r}_{p-1} + \xi(\mathbf{r}_p - \mathbf{r}_{p-1}), \quad dl = \Delta l_p d\xi. \quad (31)$$

Similarly,

$$\mathbf{r}_q(\eta) = \mathbf{r}_{q-1} + \eta(\mathbf{r}_q - \mathbf{r}_{q-1}), \quad dl' = \Delta l_q d\eta. \quad (32)$$

For ordinary non-overlapping segment-pair interactions,

$$R^{pq}(\xi, \eta) = \|\mathbf{r}_p(\xi) - \mathbf{r}_q(\eta)\|, \quad (33)$$

and

$$G_0^{pq}(\xi, \eta) = \frac{e^{-jk_0 R^{pq}(\xi, \eta)}}{4\pi R^{pq}(\xi, \eta)}. \quad (34)$$

The segment-pair contributions are [3]

$$A_{mn}^{pq} = \Delta l_p \Delta l_q \int_0^1 \int_0^1 \Lambda_m^{(p)}(\xi) \Lambda_n^{(q)}(\eta) (\hat{\mathbf{t}}_p \cdot \hat{\mathbf{t}}_q) G_0^{pq}(\xi, \eta) d\eta d\xi, \quad (35)$$

$$B_{mn}^{pq} = \Delta l_p \Delta l_q \int_0^1 \int_0^1 \Lambda_m^{(p)}(\xi) \Lambda_n^{(q)}(\eta) G_0^{pq}(\xi, \eta) d\eta d\xi. \quad (36)$$

On segment p ,

$$\Lambda_m^{(p)}(\xi) = \begin{cases} \xi, & p = m, \\ 1 - \xi, & p = m + 1, \\ 0, & \text{otherwise,} \end{cases} \quad (37)$$

and

$$\Lambda_m^{(p)} = \begin{cases} \frac{1}{\Delta l_p}, & p = m, \\ -\frac{1}{\Delta l_p}, & p = m + 1, \\ 0, & \text{otherwise.} \end{cases} \quad (38)$$

For smooth segment pairs, the double integrals in (35) and (36) are evaluated by tensor-product Gauss–Legendre quadrature. If $\{\bar{\xi}_i, \bar{w}_i\}_{i=1}^{N_g}$ are nodes and weights on $[-1, 1]$, the mapped nodes and weights on $[0, 1]$ are

$$\xi_i = \frac{\bar{\xi}_i + 1}{2}, \quad w_i = \frac{\bar{w}_i}{2}. \quad (39)$$

so,

$$\int_0^1 \int_0^1 F(\xi, \eta) d\eta d\xi \approx \sum_i \sum_j w_i w_j F(\xi_i, \eta_j). \quad (40)$$

VI. OVERLAPPING SUBSEGMENTS AND THE ψ INTEGRAL

When the supports of Λ_m and Λ_n overlap, the overlapping segment is subdivided into smaller subsegments, following the thin-wire self-term treatment used in the textbook formulation [3]. Over each small subsegment, Λ_m , Λ_n , Λ'_m , and Λ'_n are treated as constants evaluated on that subsegment. The remaining Green-function integral over a pair of subsegments is

$$\Psi_{\alpha\beta} = \int_{\Delta l_\alpha} \int_{\Delta l_\beta} G_0(\mathbf{r}, \mathbf{r}') dl' dl. \quad (41)$$

If the two subsegments are distinct and non-overlapping, $\Psi_{\alpha\beta}$ is evaluated by ordinary numerical integration. If the two subsegments overlap on the same line segment, the singular centerline interaction is replaced by the thin-wire ψ -integral [3]:

$$\psi = \int_{\Delta l} \int_{\Delta l} \frac{e^{-jk_0 R}}{4\pi R} dl' dl. \quad (42)$$

For the self subsegment, the thin-wire distance is

$$R = \sqrt{a^2 + (l - l')^2}, \quad (43)$$

where a is the wire radius. Expanding the exponential gives

$$\psi = \frac{1}{4\pi} \int_{\Delta l} \int_{\Delta l} \left[\frac{1}{R} - jk_0 - \frac{k_0^2 R}{2} + \dots \right] dl' dl. \quad (44)$$

Keeping the first two terms yields

$$\psi = \frac{\Delta l}{2\pi} \left[\ln \left(\frac{\Delta l}{a} + \sqrt{1 + \frac{(\Delta l)^2}{a^2}} \right) - \sqrt{1 + \frac{a^2}{(\Delta l)^2}} + \frac{a}{\Delta l} \right] - \frac{jk_0}{4\pi} (\Delta l)^2. \quad (45)$$

For $\Delta l \gg a$, this reduces to

$$\psi \approx \frac{\Delta l}{2\pi} \left[\ln \left(\frac{2\Delta l}{a} \right) - 1 \right] - \frac{jk_0}{4\pi} (\Delta l)^2. \quad (46)$$

In the implementation, (45) is used for same-line overlapping subsegments. It is not used for non-overlapping subsegments. It is also not used for two noncollinear segments that merely share a geometric corner. For such corner interactions, including the feed bend in the bent-wire example, refined ordinary centerline quadrature is used instead. The ψ -replacement is reserved only for same-line self subsegments.

With the subsegment approximation, the overlapping contribution to A_{mn}^{pq} is assembled as

$$A_{mn}^{pq} \approx \sum_{\alpha, \beta} \Lambda_m(l_\alpha) \Lambda_n(l_\beta) (\hat{\mathbf{t}}_p \cdot \hat{\mathbf{t}}_q) \Psi_{\alpha\beta}, \quad (47)$$

and the corresponding contribution to B_{mn}^{pq} is

$$B_{mn}^{pq} \approx \sum_{\alpha, \beta} \Lambda'_m(l_\alpha) \Lambda'_n(l_\beta) \Psi_{\alpha\beta}. \quad (48)$$

The ψ -replacement is applied only to $\Psi_{\alpha\beta}$ when $\alpha = \beta$ on the same collinear subsegment [3].

VII. DELTA-GAP EXCITATION AND INPUT IMPEDANCE

For the ideal delta-gap source, the excitation vector is assigned directly [3]:

$$V_m = \begin{cases} V_0, & m = m_f, \\ 0, & m \neq m_f, \end{cases} \quad (49)$$

where m_f is the feed basis/testing function. The matrix equation is

$$\mathbf{Z}\mathbf{I} = \mathbf{V}. \quad (50)$$

After solving for the nodal currents, the input impedance is computed using the delta-gap voltage divided by the feed current:

$$Z_{\text{in}} = \frac{V_0}{I_{m_f}}. \quad (51)$$

No magnetic-frill source is used for this project.

On segment p , with local coordinate $\xi \in [0, 1]$,

$$s = s_{p-1} + \xi \Delta l_p. \quad (52)$$

The reconstructed current is

$$I_h(s) = (1 - \xi)I_{p-1} + \xi I_p, \quad I_0 = I_{N_s} = 0. \quad (53)$$

VIII. VALIDATION GEOMETRIES

For the straight validation antenna,

$$L = 0.5\lambda, \quad a = 0.001\lambda. \quad (54)$$

The wire lies on the z -axis:

$$\mathbf{r}_i = (0, 0, z_i), \quad z_i = -\frac{L}{2} + i\frac{L}{N_s}, \quad i = 0, \dots, N_s. \quad (55)$$

An even number of physical segments is used so that the feed lies exactly on a node:

$$m_f = \frac{N_s}{2}. \quad (56)$$

The comparison impedance is

$$Z_{\text{ref}} = 73 + j42.5 \, \Omega. \quad (57)$$

This value is used as an approximate half-wave dipole benchmark rather than an exact solution, because practical and numerical dipole impedances depend on feed and thin-wire approximations [1], [3]. The relative impedance error is

$$\epsilon_Z = \frac{|Z_{\text{in}} - Z_{\text{ref}}|}{|Z_{\text{ref}}|}. \quad (58)$$

The nonparallel-wire example is a 90-degree bent wire, following the type of bent-wire validation geometry used in the textbook discussion in chapter 10 [3]. The total wire length is 2.0 m, the wire thickness is 1.0 cm, and the bend angle is 90°. Therefore,

$$L_1 = L_2 = 1.0 \, \text{m}, \quad a = 0.005 \, \text{m}. \quad (59)$$

With signed coordinate u ,

$$\mathbf{r}(u) = \begin{cases} -u \hat{\mathbf{z}}, & -L_1 \leq u \leq 0, \\ u \hat{\mathbf{x}}, & 0 \leq u \leq L_2. \end{cases} \quad (60)$$

The tangent is

$$\hat{\mathbf{t}}(u) = \begin{cases} -\hat{\mathbf{z}}, & -L_1 < u < 0, \\ \hat{\mathbf{x}}, & 0 < u < L_2. \end{cases} \quad (61)$$

For points on the same arm,

$$\hat{\mathbf{t}}(l) \cdot \hat{\mathbf{t}}(l') = 1. \quad (62)$$

For one point on the vertical arm and the other on the horizontal arm,

$$\hat{\mathbf{t}}(l) \cdot \hat{\mathbf{t}}(l') = \cos 90^\circ = 0. \quad (63)$$

Thus, the vector-potential contribution A_{mn} vanishes for cross-arm interactions. The scalar-potential contribution B_{mn} does not vanish, because it contains only the scalar Green function and basis derivatives. Hence the two arms remain coupled through B_{mn} .

The bent-wire uses $N_s = 99$ physical segments. The comparison value near 75 MHz is

$$Z_{A,\text{book}} = 41.2 - j6.6 \, \Omega. \quad (64)$$

IX. NUMERICAL RESULTS

The solver uses

$$c_0 = 2.99792458 \times 10^8 \, \text{m/s}, \quad Z_0 = 376.73031341 \, \Omega, \quad V_0 = 1 \, \text{V}. \quad (65)$$

The straight-wire study uses

$$N_s = 20, 40, 80, 160. \quad (66)$$

For each discretization, the current is reconstructed from (53), and the input impedance is computed from (51). This study checks the numerical behavior of the MoM discretization [2].

TABLE I: Main numerical parameters.

Quantity	Value	Note
c_0	2.99792458×10^8	m/s
Z_0	376.73031341	Ω
V_0	1	V
Straight L	0.5λ	required case
Straight a	0.001λ	required case
Straight N_s	20, 40, 80, 160	discr. study
Bent L	2.0 m	nonparallel ex.
Bent a	0.005 m	radius
Bent $2a$	1.0 cm	diameter
Bent N_s	99	bent-wire

The straight-dipole current distributions show the expected endpoint zeros and a feed-centered current maximum. The normalized current shape remains close to the expected half-wave profile as the discretization is refined, which is consistent with the standard qualitative behavior of center-fed thin dipoles [1]. The input-impedance plot shows the real and imaginary parts as functions of segment count, and the error plot compares the computed impedance with $73 + j42.5 \, \Omega$.

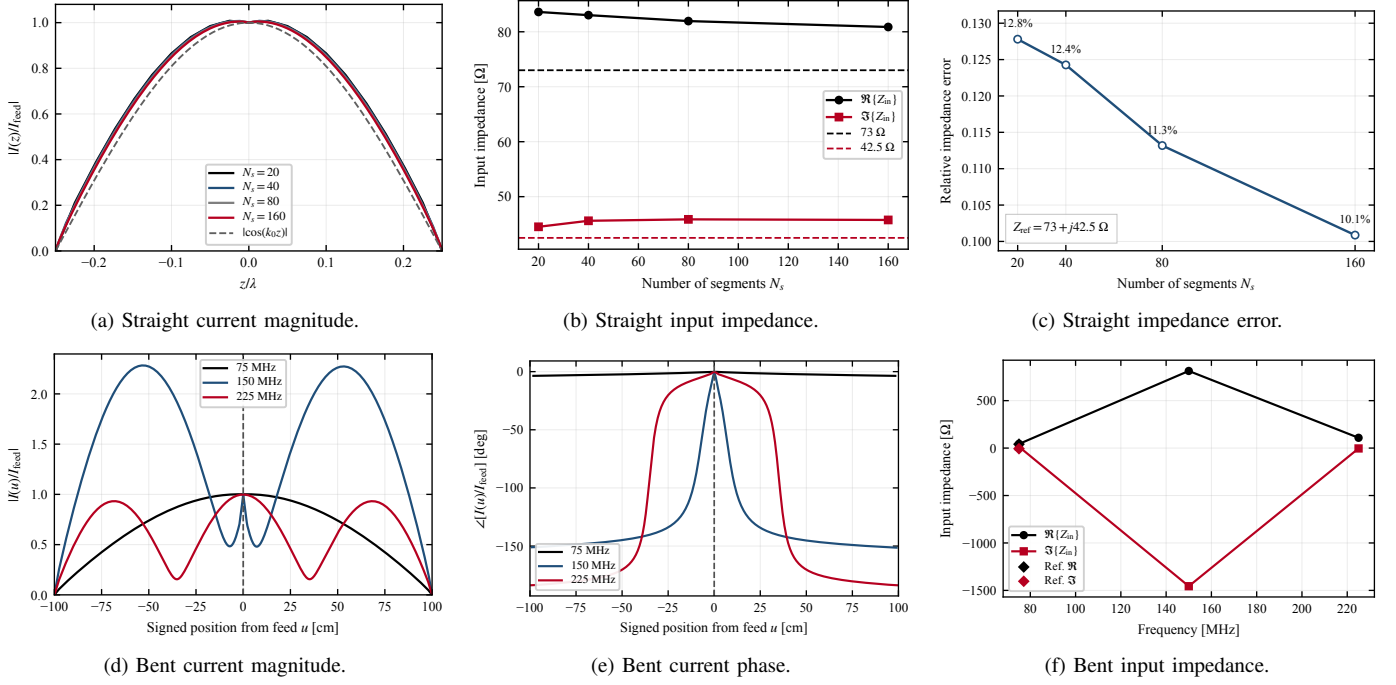


Fig. 1: Numerical results for the straight half-wave dipole and the 90-degree bent-wire antenna. The straight-wire plots show current magnitude, impedance convergence, and relative error. The bent-wire plots show current magnitude, current phase, and input impedance.

TABLE II: Straight half-wave dipole input impedance and numerical checks.

N_s	$\Re\{Z_{in}\}$	$\Im\{Z_{in}\}$	ϵ_Z	Res.
20	83.6057	44.5156	1.278×10^{-1}	8.45×10^{-15}
40	83.0257	45.6096	1.243×10^{-1}	3.69×10^{-14}
80	81.9468	45.8704	1.132×10^{-1}	1.59×10^{-13}
160	80.8714	45.7622	1.009×10^{-1}	5.57×10^{-13}

TABLE III: Ninety-degree bent-wire input impedance and numerical checks.

f MHz	$\Re\{Z_{in}\}$ Ω	$\Im\{Z_{in}\}$ Ω	Res.
75	43.3965	13.9469	7.44×10^{-13}
150	812.8377	-1455.8369	2.07×10^{-14}
225	108.4299	-3.8664	7.28×10^{-14}

The computed straight-dipole impedances and numerical residuals are summarized in Table II. The matrix residual remains near machine precision for all discretizations. The computed values are within about 10% of the approximate reference value. The remaining difference is consistent with the sensitivity of the delta-gap thin-wire model to basis order, self-term approximation, segment aspect ratio, and feed modeling [1], [3].

The bent-wire numerical values are listed in Table III. The resistance at 75 MHz is close to the quoted resistance, while the reactance differs because the bent-wire feed region is sensitive to local feed and corner treatment, including self interactions, adjacent-segment coupling, near-singular quadrature, and finite-radius regularization. [3].

At 75 MHz, the computed bent-wire impedance is

$$Z_{in} = 43.3965 + j13.9469 \Omega.$$

The computed input impedance at 75 MHz is compared with the quoted reference value $41.2 - j6.6 \Omega$. The resistance is in reasonable agreement, while the reactance shows a larger difference.

X. CONCLUSION

The MoM solver produced physically consistent current distributions and input impedances for both straight and bent thin-wire antennas. For the half-wave dipole, discretization refinement gave stable solutions and reasonable agreement with the approximate $73 + j42.5 \Omega$ reference. For the bent-wire case, the solver captured the effect of nonparallel segment coupling and produced a resistance close to the quoted comparison value at 75 MHz. Future work could focus on improved feed modeling, higher-order bases, and more accurate near-singular quadrature.

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